L2/L3 HH Warm-Start PAOP+HVA Results (Human-Readable PDF Report)

Part I - L=3 Results

1. Parameter Manifest

- Model family/name: `Hubbard` (HH specialization).
- Ansatz types covered: `hh_hva_tw`, `hh_hva_ptw`, ADAPT(`paop_lf_std`), ADAPT trend pools (`uccsd+paop`, `uccsd+paop+hva`).
- Drive enabled: both `false` (static sections) and `true` (driven section from `drive_from_fix1_warm_start_B_full.json`).
- Core physical parameters: Part I (L3) uses `t=1.0, U=4.0, dv=0.0`; Part II (L2 analog) also uses `t=1.0, U=4.0, dv=0.0`.
- HH-defining parameters for L3/L2 analog sections: `omega0=1.0`, `g_ep=0.5`, `n_ph_max=1`, boson encoding `binary`, ordering `blocked`, boundary `open`.
- Legacy baseline note: the `L2 VQE` row retained in Part I scoreboard is historical provenance at `U=2.0`, `g_ep=1.0`.
- Reproducibility sector references: L2 `(N_up,N_dn)=(1,1)`; L3 `(N_up,N_dn)=(2,1)`.
- Driven-run timing/drive: L3 uses `t_final=15.0`, `num_times=201`, `trotter_steps=192`, `exact_steps=384`; L2 analog uses `t_final=15.0`, `num_times=201`, `trotter_steps=128`, `exact_steps=256`; both use `A=0.5`, `omega=2.0`, `phi=0.0`, `t0=0.0`, `tbar=3.0`, pattern `staggered`.

1.1 What "Accessibility Rerun" Means (Exhaustive Definition)

In this report, "Accessibility rerun" does **not** mean a UI/accessibility feature and does **not** mean driven Trotter dynamics. It means a second-pass static ADAPT optimization run where we keep a fixed practical compute budget and re-run the warm-start branch under those fixed limits to measure what quality is realistically reachable.

Concretely, the accessibility runs come from `ACC_JSON = artifacts/useful/L3/l3_hh_accessibility_fixes_under8pct.js` Each rung (for example, `Acc B` and `Acc C`) specifies explicit ADAPT limits such as `adapt_max_depth`, `adapt_maxiter`, `eps_grad`, and `wallclock_cap_s=1200`. The purpose is to compare outcomes under controlled run limits, not to change physics definitions.

The non-accessibility counterpart in this PDF is `B export` from `B_EXPORT_JSON = artifacts/useful/L3/warmstart_states/` which is the direct exported rebuild artifact for warm-start branch B. That is why `B export` and `Acc B` can be very close in `DeltaE` but still differ in recorded depth/parameter count (`42` vs `43`) and runtime fields: they are separate executions with separate bookkeeping.

For this reason, throughout this document: - `Accessibility rerun` = fixed-budget static ADAPT rerun (`Acc B`, `Acc C`). - `Driven dynamics` = time-evolution branch analysis from `drive_from_fix1_warm_start_B_full.json`.

2. Executive Summary

- L2 strong VQE reaches `DeltaE=6.029e-07` (near exact).
- L3 warm VQE alone remains at `DeltaE=1.969e-02`.
- Warm-start + ADAPT improves L3 to `DeltaE=4.393e-03` (Accessibility C).
- Trend runs reach `DeltaE~2.622e-04` (best shown point).
- Drive branch diagnostics show PAOP branch fidelity near `0.995893` vs HVA near `0.827750` against the same filtered exact manifold.

3. Math and Metrics (Compact)

$$H = H_t + H_U + H_{ph} + H_{e-ph}, \quad H(t) = H + H_{drive}(t).$$

$$E_{exact,sector} = \min_{\psi \in \mathcal{H}_{(N_\uparrow, N_\downarrow)}} \langle \psi | H | \psi \rangle.$$

$$\Delta E = |E_{best} - E_{exact,sector}|, \quad \varepsilon_{rel} = \Delta E / |E_{exact,sector}|.$$

$$|\psi_d\rangle = e^{-i\theta_d G_d} \cdots e^{-i\theta_1 G_1}|\psi_0\rangle, \quad g_m = i\langle\psi_d|[H, G_m]|\psi_d\rangle.$$

Cost metrics used in this report:

$$\kappa_{eval} = nfev/P, \quad \tau_{eval} = runtime_s/nfev.$$

4. Run Labels and Measurement Dictionary

4.1 Run Label Glossary (What each row name means)

Label	Short definition
`L2 VQE`	Hardcoded HH VQE on L=2 (`hh_hva_tw`, `L-BFGS-B`).
`L3 warm VQE`	Warm-start seed VQE stage on L=3 (`hh_hva_ptw`, `COBYLA`).
`B export`	Exported fix1 warm-start branch B state (`depth=42`, `pool=paop_lf_std`).
`Acc B`	Accessibility rerun branch B (separate execution from `B export`, reached `depth=43`).
`Acc C`	Accessibility rerun branch C (different rung settings).
`A_medium`	Trend pool A (`uccsd+paop`), medium budget (`depth=20`).
`A_heavy`	Trend pool A (`uccsd+paop`), heavy budget (`depth=36`).
`B_medium`	Trend pool B (`uccsd+paop+hva`), medium budget (`depth=20`).
`B_heavy`	Trend pool B (`uccsd+paop+hva`), heavy budget (`depth=36`).

Provenance key: - `L2_JSON` = `artifacts/useful/L2/H_L2_hh_termwise_regular_lbfgs_t1.0_U2.0_g1_nph1.json` - `L3_WARM_JSON` = `artifacts/json/hh_L3_hh_hva_ptw_heavy.json` - `B_EXPORT_JSON` = `artifacts/useful/L3/warmstart_s` - `ACC_JSON` = `artifacts/useful/L3/l3_hh_accessibility_fixes_under8pct.json` - `TREND_JSON` = `artifacts/useful/L3/l3_uccsd_paop_hva_trend_full_20260302T000521.json`

Clarification on naming: - In this report, `Acc B` means "Accessibility rung B" (same concept you referred to as "BACC"). - `B export` and `Acc B` are related but not identical runs; that is why depth can appear as `42` vs `43`.

4.2 Measurement Dictionary (What each metric is)

Core optimization metrics:

Metric	Definition	Units
`E_best`	`vqe.energy` or `result.E_best`	energy
`E_exact`	`ground_state.exact_energy_filtered` or `exact.E_exact_sector`	energy
`DeltaE`	`|E_best - E_exact|`	energy
`eps_rel`	`DeltaE / |E_exact|`	unitless
`Params P`	`vqe.num_parameters` or `result.num_parameters`	count
`nfev`	`vqe.nfev` or `result.nfev_total`	count
`Depth`	VQE proxy=`reps`; ADAPT depth = `ansatz_depth` or `adapt_depth_reached`	count
`Runtime (s)`	`runtime_s` (or nearest runtime field)	seconds

Metric	Definition	Units
`kappa_eval`	`nfev / P`	eval/param
`tau_eval`	`runtime_s / nfev`	s/eval

Dynamics metrics:

Metric	Definition	Units
`F_paop` / `F_hva`	`fidelity_paop_trotter`, `fidelity_hva_trotter`	unitless
`dE_total_paop` / `dE_total_hva`	`|E_total_exact - E_total_trotter|` per branch	energy
`dD_paop` / `dD_hva`	`|D_exact - D_trotter|` per branch	unitless

5. Readable Static Scoreboard

Run	DeltaE	Depth	Params P	nfev	E_best	E_exact	eps_rel	Optimizer
L2 VQE	6.029e-07	6	108	15042	-0.389552500	-0.389553103	1.548e-06	L-BFGS-B
L3 warm VQE	1.969e-02	3	39	4000	0.264627255	0.244940700	8.037e-02	COBYLA
B export	6.508e-03	42	42	n/a	0.251448234	0.244940700	2.657e-02	ADAPT
Acc B	6.508e-03	43	43	8063	0.251448234	0.244940700	2.657e-02	ADAPT
Acc C	4.393e-03	38	38	6227	0.249334000	0.244940700	1.794e-02	ADAPT
A_medium	2.622e-04	20	20	12640	0.245202940	0.244940700	1.071e-03	ADAPT
A_heavy	2.629e-04	36	36	26833	0.245203648	0.244940700	1.074e-03	ADAPT
B_medium	2.622e-04	20	20	12640	0.245202940	0.244940700	1.071e-03	ADAPT
B_heavy	2.629e-04	36	36	26833	0.245203648	0.244940700	1.074e-03	ADAPT

Note: this table intentionally rounds values for readability; all source values are retained in JSON artifacts.

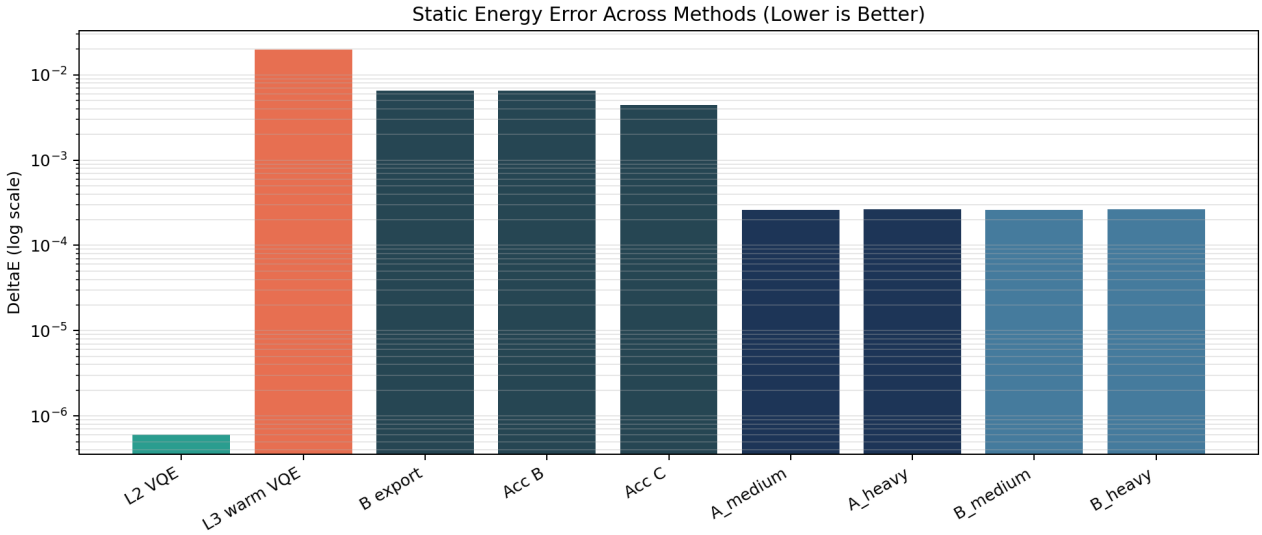

Figure 1: Static error comparison

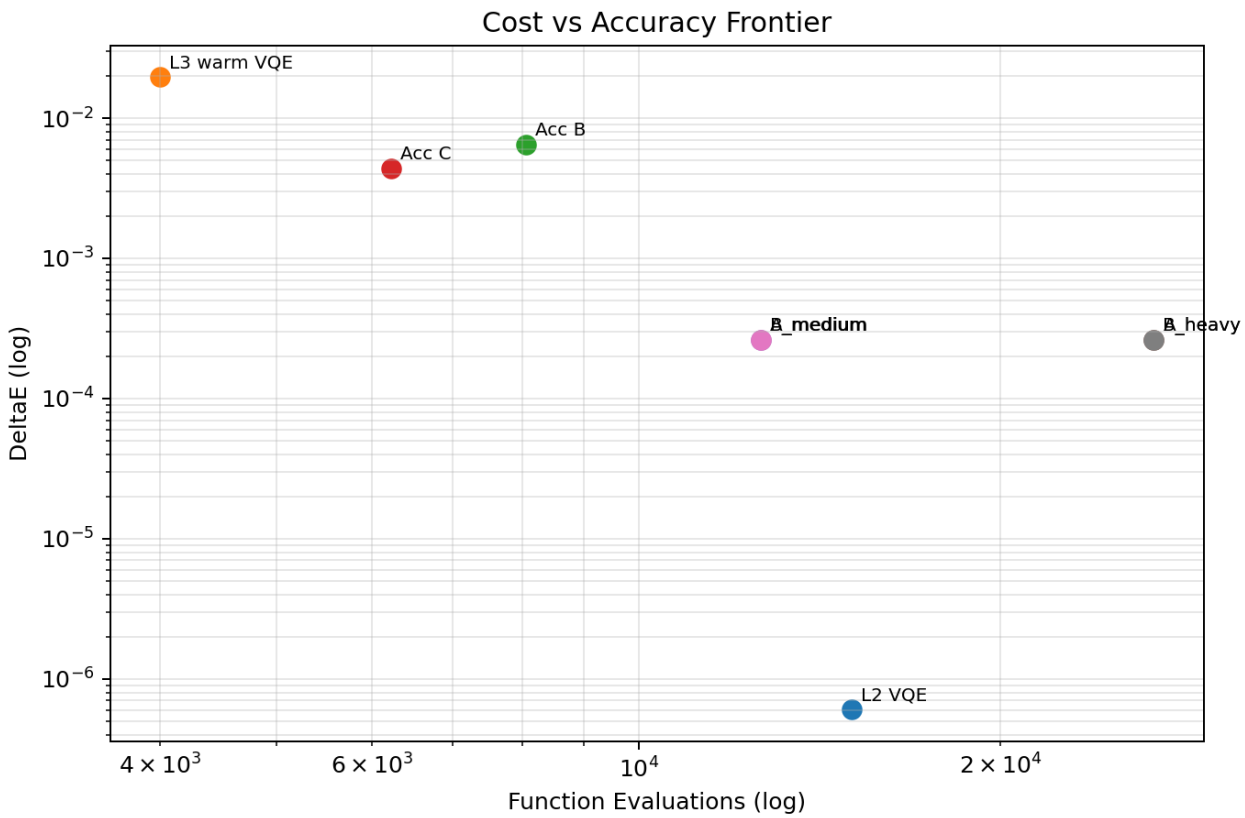

Figure 2: Cost vs accuracy

6. Accessibility and Warm-Start Convergence

Accuracy/cost view:

Run	DeltaE_best	Depth	Params	nfev	Runtime (s)
B export	6.508e-03	42	42	n/a	1201.4
Acc B	6.508e-03	43	43	8063	1261.9
Acc C	4.393e-03	38	38	6227	1208.3

Warm-start gain view:

Run	DeltaE_warm	DeltaE_best	Gain factor (`DeltaE_warm/DeltaE_best`)	E_warm	E_best
B export	1.969e-02	6.508e-03	3.025	0.264627255	0.251448234
Acc B	1.969e-02	6.508e-03	3.025	0.264627255	0.251448234
Acc C	1.234e-02	4.393e-03	2.809	0.257282637	0.249334000

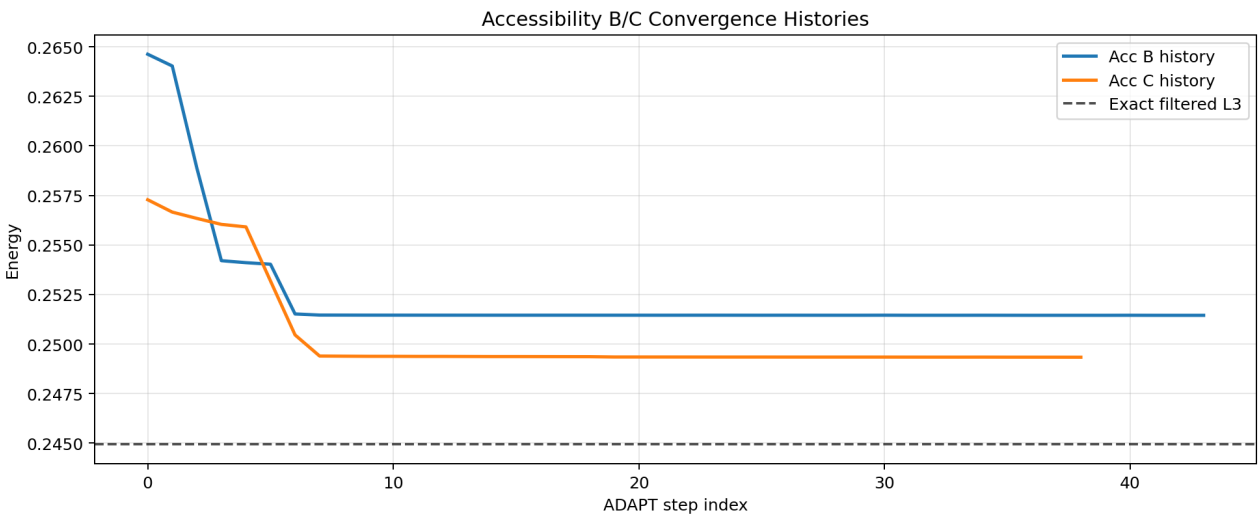

Figure 3: Accessibility convergence histories

6.1 Operator Family Symbol Definitions (Figure 4)

The three family labels in Figure 4 correspond to the PAOP generators:

$$P_i = i(b_i^\dagger - b_i), \quad n_i = n_{i\uparrow} + n_{i\downarrow}, \quad \tilde{n}_i = n_i - \bar{n}.$$

$$K_{ij} = \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma}\right), \quad J_{ij} = i \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(c_{i\sigma}^\dagger c_{j\sigma} - c_{j\sigma}^\dagger c_{i\sigma}\right).$$

- `Disp`: $G_i^{\text{disp}} = \tilde{n}_i P_i$
- `Hop Drag`: $G_{ij}^{\text{hopdrag}} = K_{ij}(P_i - P_j)$
- `Cur Drag`: $G_{ij}^{\text{curdrag}} = J_{ij}(P_i - P_j)$

`Cur Drag` is the same `paop_curdrag` family (current-drag channel); this is the channel you referred to as Kerr-drag in discussion.

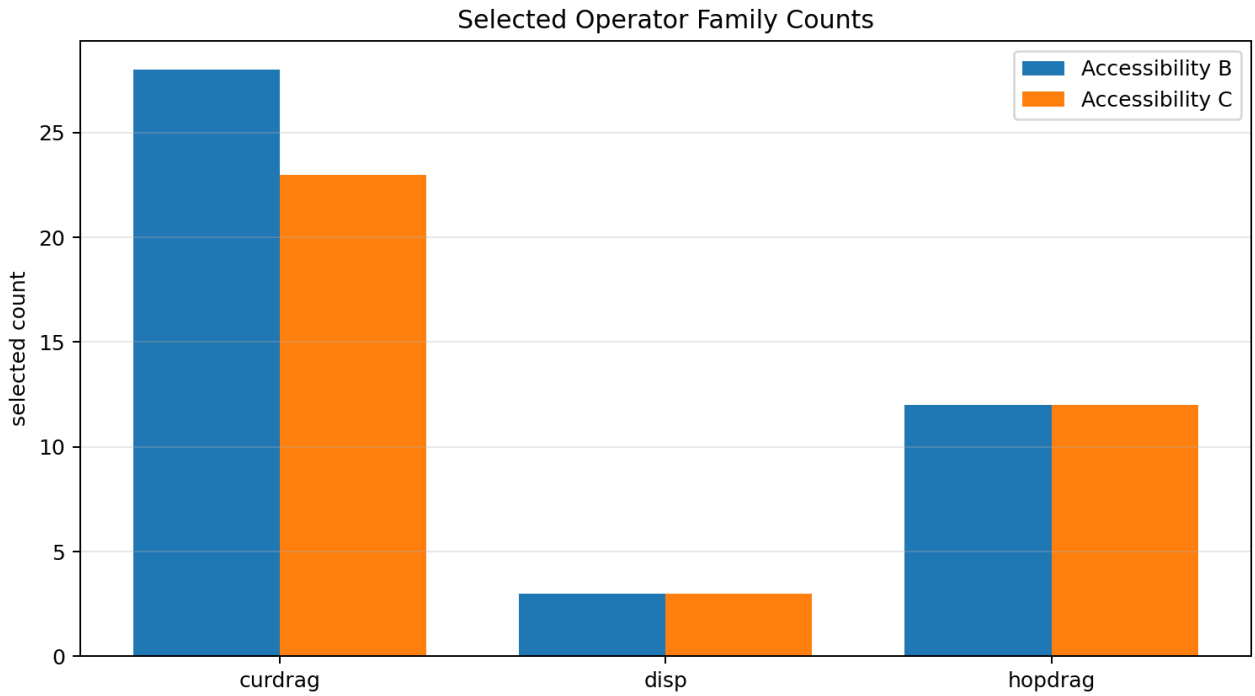

Figure 4: Selected operator family counts

7. Trend Experiment (A/B pools)

- Raw pool sizes: `UCCSD=8`, `PAOP=7`, `HVA=13`.
- Pool A (`uccsd+paop`): `dedup_total=15`, `overlap_count=0`.
- Pool B (`uccsd+paop+hva`): `dedup_total=28`, `overlap_count=0`.

Trend run	DeltaE	Depth	Params P	nfev	runtime (s)	final max
A_medium	2.622e-04	20	20	12640	624.0	4.564e-03
A_heavy	2.629e-04	36	36	26833	1722.5	7.722e-03
B_medium	2.622e-04	20	20	12640	720.7	4.564e-03
B_heavy	2.629e-04	36	36	26833	2379.3	7.722e-03

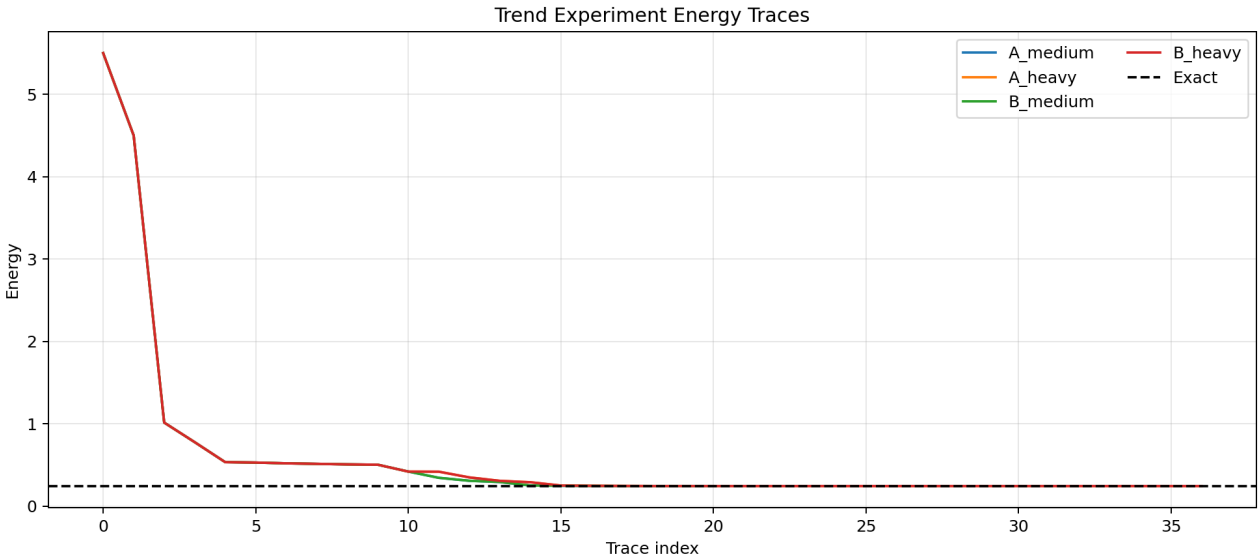

Figure 5: Trend energy traces

Section role boundary: Section 7 is static ADAPT trend optimization; Section 8 is driven branch dynamics.

8. Drive Dynamics (Branch-Aware)

Branch order: `exact_gs_filtered`, `exact_paop`, `trotter_paop`, `exact_hva`, `trotter_hva`.

8.0 Dynamics Reference Trajectory and How It Is Generated

For dynamics, the reference is the time-dependent exact trajectory

$$E_{ref}(t) \equiv E_{exact_gs_filtered}(t) = texttt energy_t otal_e xact(t),$$

taken from the `exact_gs_filtered` branch in `drive_from_fix1_warm_start_B_full.json`.

This is not a single static scalar; it is a full time series used as the baseline for branch comparisons. In this run: - `E_ref(t=0) = 0.24494070012791422` - `E_ref(t=t_final=15) = 0.26953974487978255`

Repo options/fields used to generate this dynamics dataset (same file, `settings.drive` + top-level settings): - drive enabled: `true` - waveform: `A=0.5`, `omega=2.0`, `phi=0.0`, `t0=0.0`, `tbar=3.0`, pattern=`staggered` - Trotter setup: `suzuki_order=2`, `trotter_steps=192` - exact reference setup: `reference_steps_multiplier=2`, `reference_steps=384` - exact reference method: `exponential_midpoint_magnus2_order2` - backend: `scipy_sparse_expm_multiply`

8.1 Exact-GS vs PAOP/Trotter-PAOP (Primary Dynamics Comparison)

Naming map used in this section (`drive_from_fix1_warm_start_B_full.json`): - `exact_gs_filtered`: exact/reference propagation from filtered-sector exact ground-state initialization. - `exact_paop`: exact propagation from imported PAOP ADAPT state (`source=adapt_json`, `pool=paop_lf_std`, `depth=42`). - `trotter_paop`: Trotter propagation from the same PAOP initialization. - `exact_hva`: exact propagation from regular hardcoded VQE (`source=regular_vqe`, `ansatz=hh_hva_ptw`). - `trotter_hva`: Trotter propagation from the same HVA initialization.

Primary PAOP triad (total-energy distances):

Metric pair	t=0	mean over time	max over time	t=t_final
`|E_gs - E_exact_paop|`	6.508e-03	6.941e-03	7.938e-03	6.923e-03
`|E_gs - E_trotter_paop|`	6.508e-03	6.936e-03	7.910e-03	6.789e-03
`|E_exact_paop - E_trotter_paop|`	0.000e+00	1.380e-04	1.014e-03	1.340e-04

Companion HVA comparison (same metric form):

Metric pair	t=0	mean over time	max over time	t=t_final
`|E_gs - E_exact_hva|`	1.777e-01	1.859e-01	1.905e-01	1.876e-01
`|E_gs - E_trotter_hva|`	1.777e-01	1.859e-01	1.904e-01	1.868e-01
`|E_exact_hva - E_trotter_hva|`	0.000e+00	4.360e-04	2.766e-03	7.359e-04

Fidelity summary (vs filtered exact GS manifold projector):

Branch	mean	min	max	final
`paop`	0.995893	0.995732	0.996043	0.995881
`hva`	0.827750	0.827244	0.828695	0.828071

Driven overlays requested in this revision: - Plot A combines absolute energies and energy-error trajectories in one figure. - Plot B combines absolute energies, fidelities, and observables in one figure.

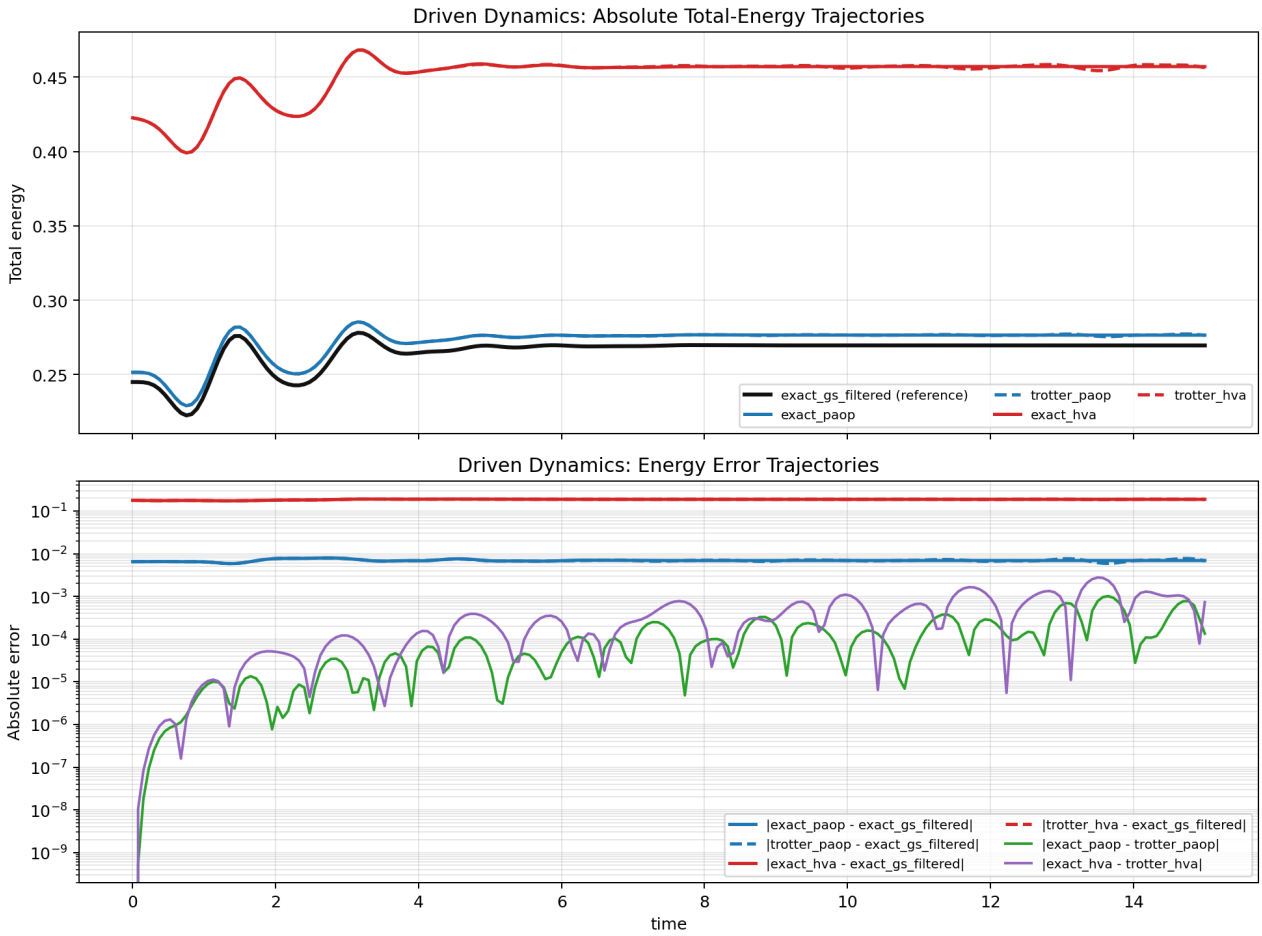

Figure 7: Drive absolute energies + energy errors

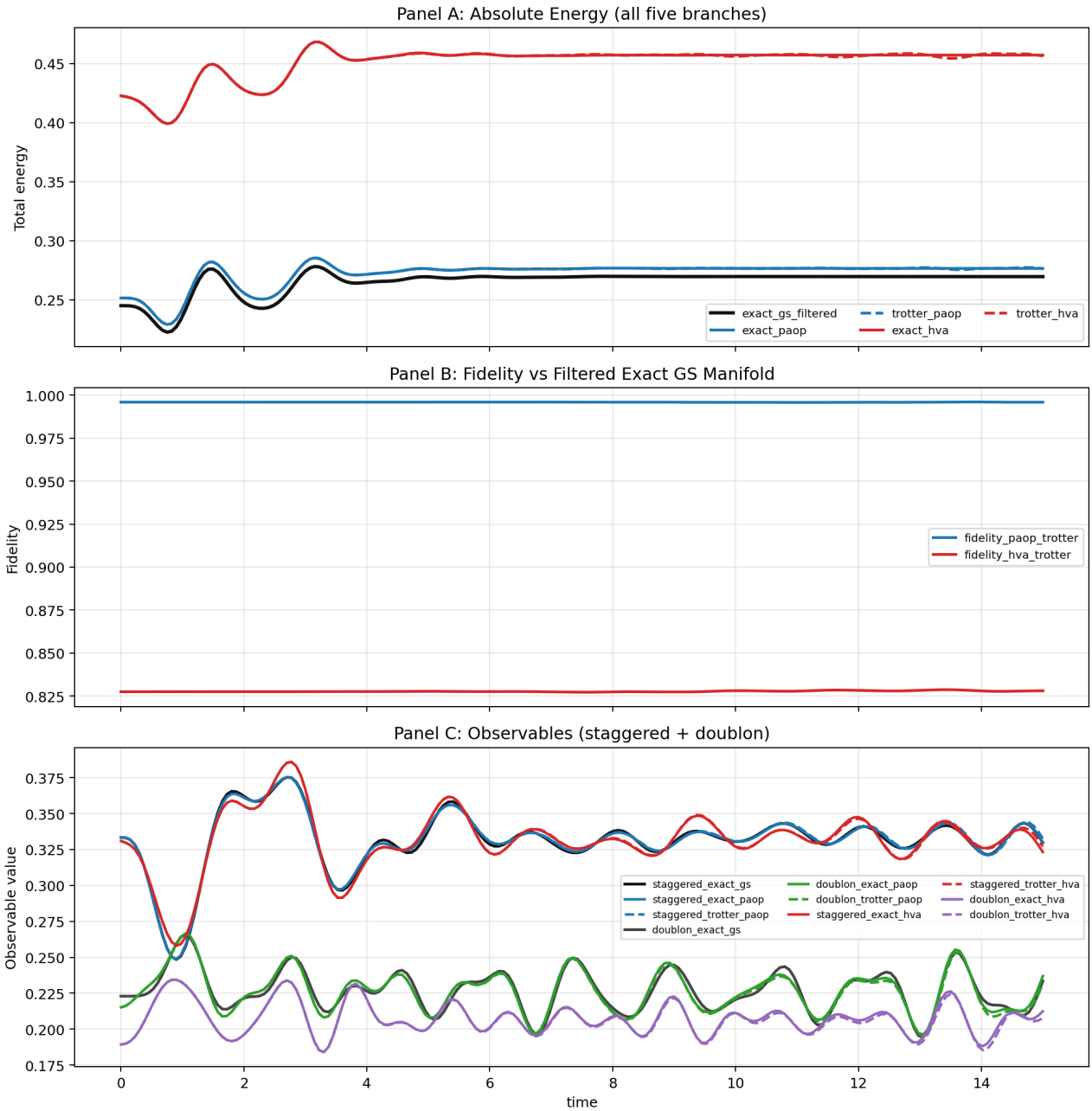

Figure 8: Drive energies + fidelities + observables

8.2 L=3 3D Energy Panels (Static vs Driven)

Static-energy 3D panel from the L3 drive artifact (`energy_static_*` lanes; this is an observable view, not a separate no-drive run). Source JSON: `artifacts/useful/L3/drive_from_fix1_warm_start_B_full.json`.

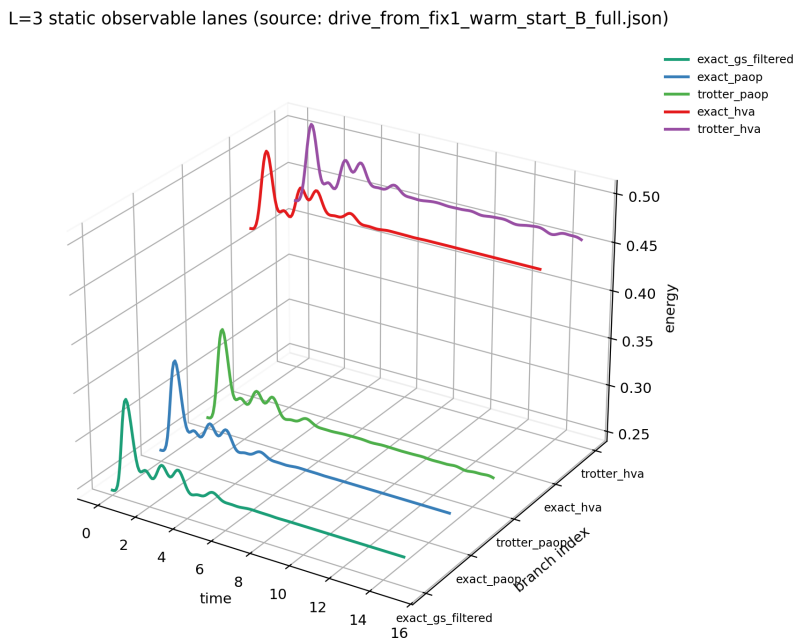

Figure 9: L3 3D static-energy lanes

Driven total-energy 3D panel from the same artifact (`energy_total_*` lanes). Source JSON: `artifacts/useful/L3/drive_fro`

8.3 Additional Branch Aggregates

Metric	PAOP branch	HVA branch
Fidelity mean	0.995892975	0.827750111
Fidelity final	0.995881343	0.828070675
max	Delta E_total	
mean	Delta E_total	
max	Delta doublon	

9. Interpretable Time-Snapshot Appendix (No Raw Dump)

Snapshot rows every 10th time index for readability (21 rows total).

idx	time	F_paop	F_hva	dE_total_paop	dE_total_hva	dD_paop	dD_hva
0	0.000	0.995909	0.827522	0.000e+00	0.000e+00	0.000e+00	0.000e+00
10	0.750	0.995909	0.827521	1.663e-06	1.320e-06	9.340e-06	1.044e-05
20	1.500	0.995909	0.827517	7.830e-06	1.727e-05	7.975e-06	9.208e-06
30	2.250	0.995904	0.827514	6.163e-06	3.898e-05	5.051e-05	5.763e-05
40	3.000	0.995906	0.827580	1.832e-05	1.211e-04	7.286e-05	8.990e-05

idx	time	F_paop	F_hva	dE_total_paop	dE_total_hva	dD_paop	dD_hva
50	3.750	0.995910	0.827601	4.004e-05	4.533e-05	1.343e-04	8.513e-05
60	4.500	0.995910	0.827710	6.052e-05	2.118e-04	2.869e-04	2.994e-04
70	5.250	0.995936	0.827729	1.259e-05	8.408e-05	3.732e-04	3.713e-04
80	6.000	0.995924	0.827590	4.948e-05	2.581e-04	4.923e-04	4.956e-04
90	6.750	0.995951	0.827563	1.030e-04	1.305e-04	3.830e-04	4.241e-04
100	7.500	0.995918	0.827248	1.660e-04	6.827e-04	3.472e-04	3.184e-04
110	8.250	0.995908	0.827536	9.220e-05	8.233e-05	4.452e-04	1.238e-04
120	9.000	0.995863	0.827396	2.033e-04	3.001e-04	7.400e-04	9.808e-04
130	9.750	0.995822	0.827926	1.248e-04	5.640e-04	9.674e-04	1.164e-03
140	10.500	0.995833	0.827907	9.978e-05	1.265e-04	1.854e-03	2.196e-03
150	11.250	0.995769	0.828147	3.302e-04	1.730e-04	6.147e-04	9.901e-04
160	12.000	0.995880	0.828300	2.780e-04	9.137e-04	1.374e-03	1.826e-03
170	12.750	0.995851	0.828023	4.194e-05	1.302e-03	1.309e-04	3.459e-04
180	13.500	0.995954	0.828678	7.711e-04	2.766e-03	3.432e-03	1.780e-03
190	14.250	0.995912	0.827734	1.078e-04	1.247e-03	3.170e-03	3.969e-03
200	15.000	0.995881	0.828071	1.340e-04	7.359e-04	5.556e-03	4.733e-03

10. Quantile Appendix (Trajectory Error Distributions)

quantile	dE_total_paop	dE_total_hva	dD_paop	dD_hva
min	0.000e+00	0.000e+00	0.000e+00	0.000e+00
q25	1.345e-05	4.970e-05	9.817e-05	8.990e-05
median	6.784e-05	2.426e-04	4.109e-04	3.386e-04
q75	1.702e-04	6.503e-04	1.030e-03	9.851e-04
q90	3.296e-04	1.141e-03	1.963e-03	2.055e-03
q99	9.172e-04	2.538e-03	4.475e-03	4.095e-03
max	1.014e-03	2.766e-03	5.556e-03	4.733e-03

Part II - L=2 Analog Run (New)

11. L=2 Analog Run (New)

11.1 Provenance

- Exported warm-start state:
  - `artifacts/useful/L2/warmstart_states/fix1_warm_start_B_l2_state.json`
- Static L2 analog run:
  - `artifacts/useful/L2/l2_open_blocked_static_from_fix1B.json`
- Driven L2 analog run:
  - `artifacts/useful/L2/l2_open_blocked_drive_from_fix1B.json`
- Strict trend crosscheck:
  - `artifacts/useful/L2/l2_uccsd_paop_hva_trend_crosscheck.json`

11.2 L2 Static Scoreboard (DeltaE-first ordering)

Run	DeltaE	Depth	Params	nfev	Stop/Notes
L2 accessibility export (`fix1_warm_start_C`)	1.087e-03	78	78	36889	`wallclock_cap`

Run	DeltaE	Depth	Params	nfev	Stop/Notes
L2 internal HVA VQE (static pipeline)	8.056e-03	2	16	1200	`hh_hva_ptw`, `COBYLA`
L2 trend crosscheck (`A_heavy`)	3.283e-01	6	6	264	`eps_energy`

11.3 L2 Drive Dynamics Triad (same format as L3)

Primary PAOP triad (static-energy distances in driven run):

Metric pair	t=0	mean over time	max over time	t=t_final
`|E_gs - E_exact_paop|`	1.102e-03	1.294e-03	2.007e-03	1.262e-03
`|E_gs - E_trotter_paop|`	1.102e-03	1.857e-03	4.761e-03	1.561e-03
`|E_exact_paop - E_trotter_paop|`	0.000e+00	7.150e-04	3.500e-03	2.995e-04

Companion HVA comparison:

Metric pair	t=0	mean over time	max over time	t=t_final
`|E_gs - E_exact_hva|`	8.056e-03	8.942e-03	2.228e-02	8.452e-03
`|E_gs - E_trotter_hva|`	8.056e-03	9.691e-03	2.220e-02	1.057e-02
`|E_exact_hva - E_trotter_hva|`	0.000e+00	9.449e-04	5.423e-03	2.115e-03

L2 fidelity summary:

Branch	mean	min	max	final
`paop`	0.999196	0.998322	0.999389	0.998603
`hva`	0.998070	0.997306	0.998337	0.997497

11.4 L2 3D Energy Panels (Static vs Driven)

All panels in this subsection are generated from L2 JSON only (no L3 trajectory source is used here).

L2 static no-drive 3D energy lanes. Source JSON: `artifacts/useful/L2/l2_open_blocked_static_from_fix1B.json`.

L2 driven total-energy 3D lanes. Source JSON: `artifacts/useful/L2/l2_open_blocked_drive_from_fix1B.json`.

11.5 L2 Operator Family Counts (Cur-Drag/Kerr-Drag Consistency)

L2 includes the same family labels as L3 (`disp`, `hopdrag`, `curdrag`), with `curdrag` explicitly present.

L2 accessibility run	Disp	Hop Drag	Cur Drag	Total selected depth
`fix1_warm_start_B`	2	5	62	69
`fix1_warm_start_C`	4	73	1	78

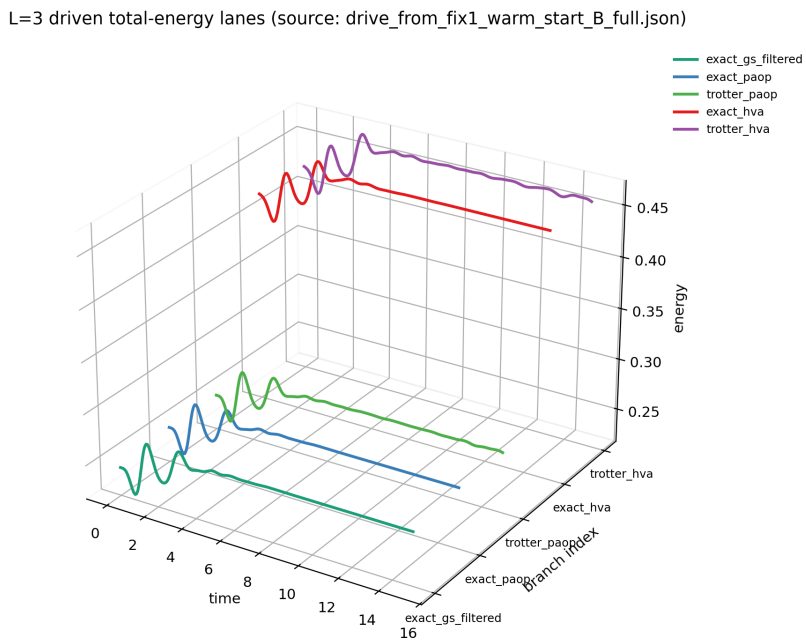

Figure 10: L3 3D driven total-energy lanes

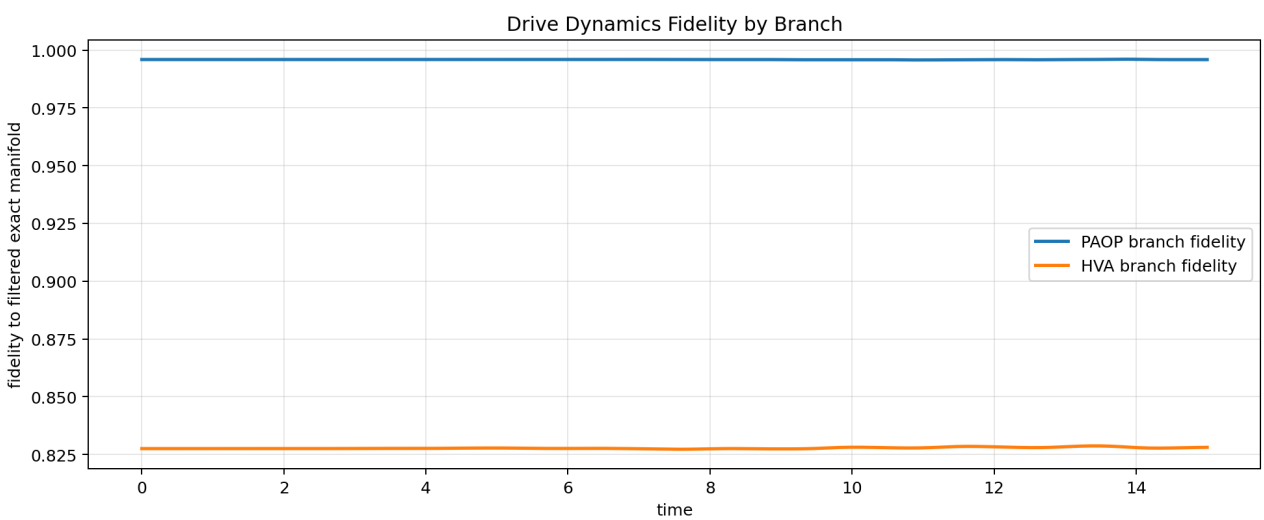

Figure 11: Drive fidelity

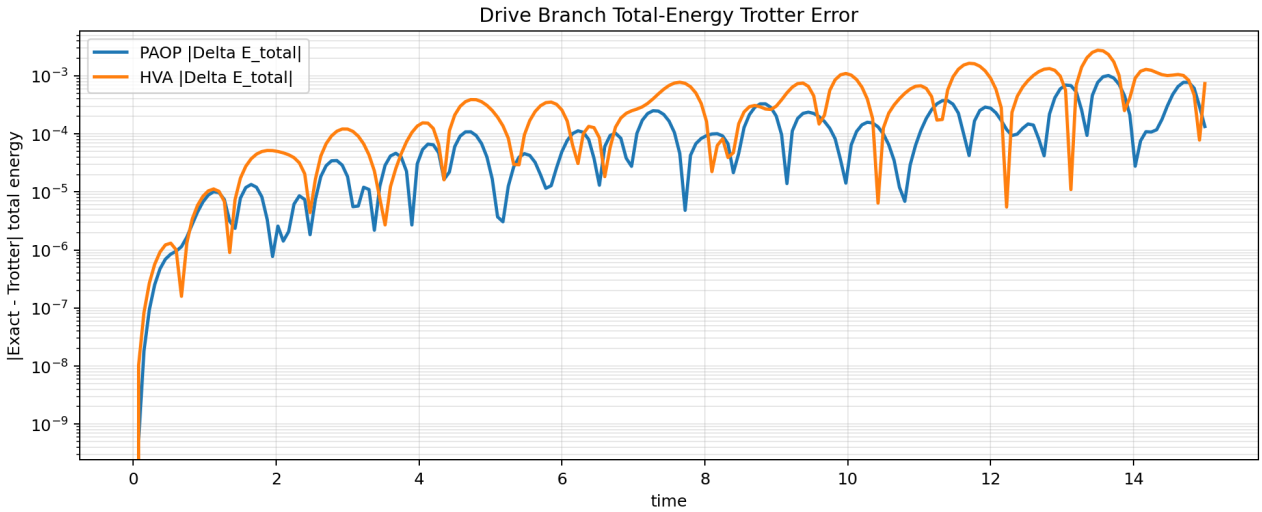

Figure 12: Drive total-energy error

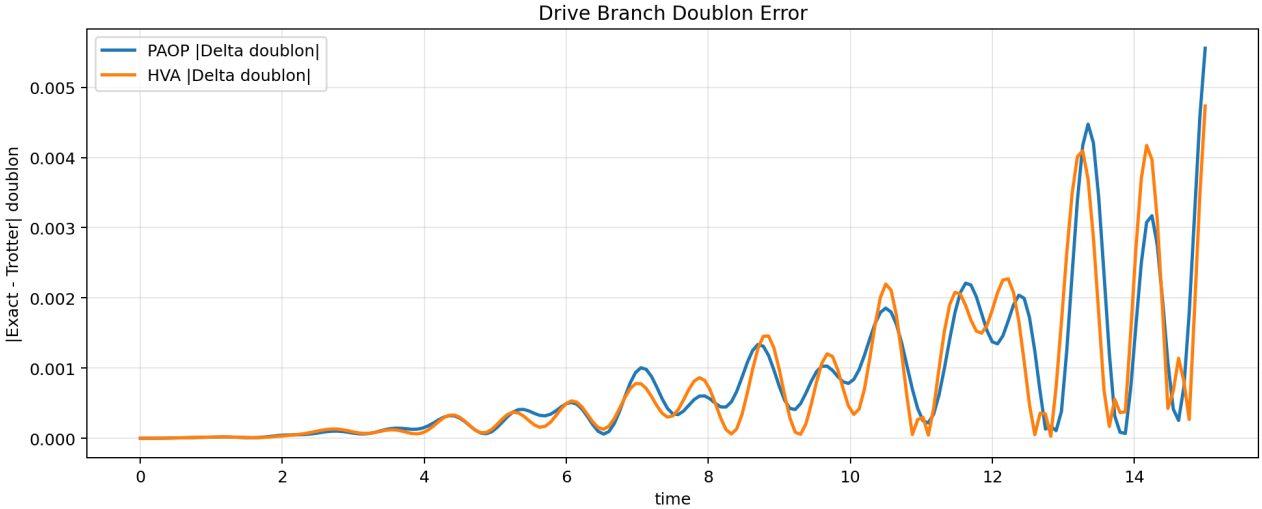

Figure 13: Drive doublon error

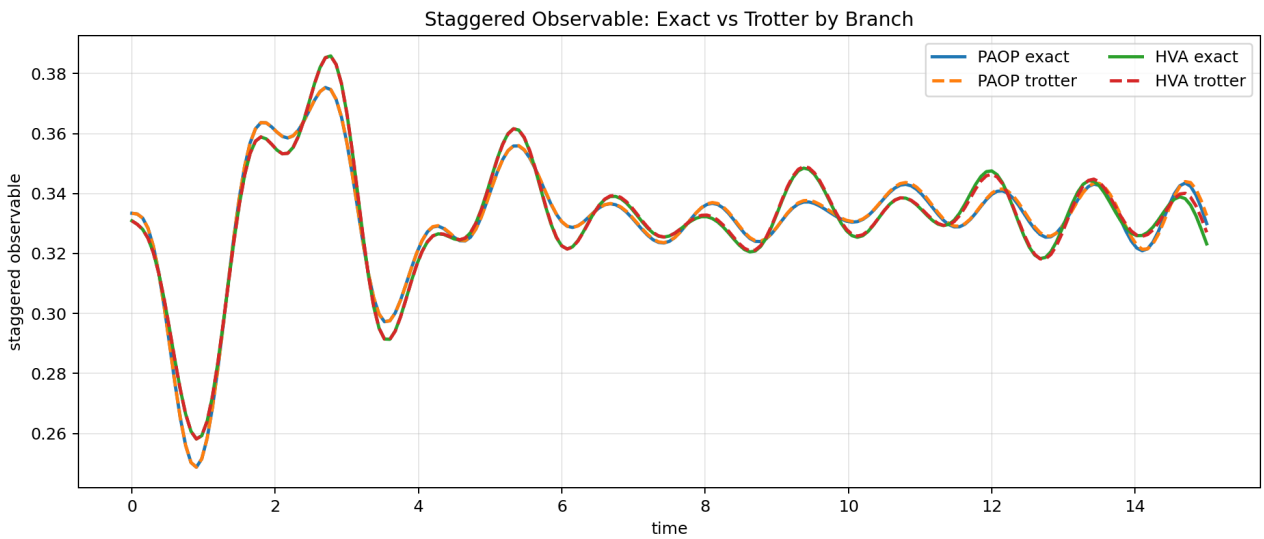

Figure 14: Staggered observable overlay

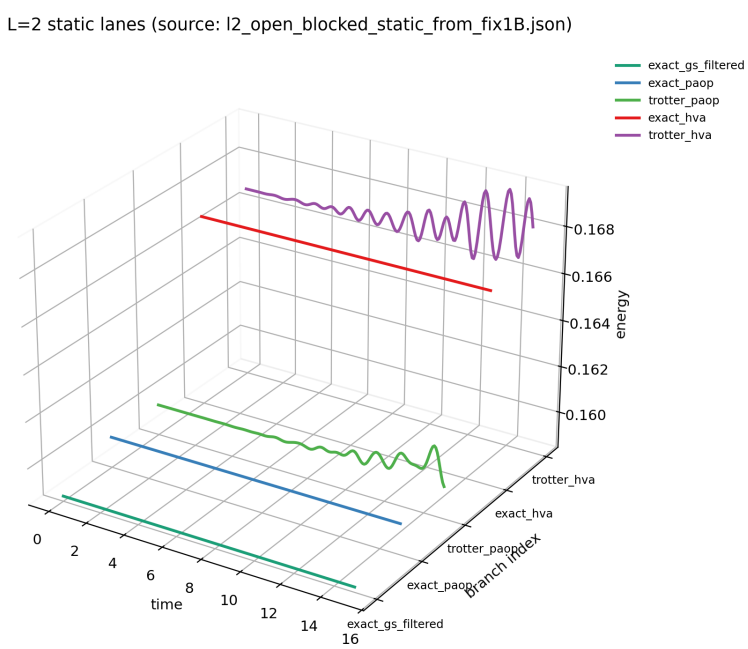

Figure 15: L2 3D static-energy lanes

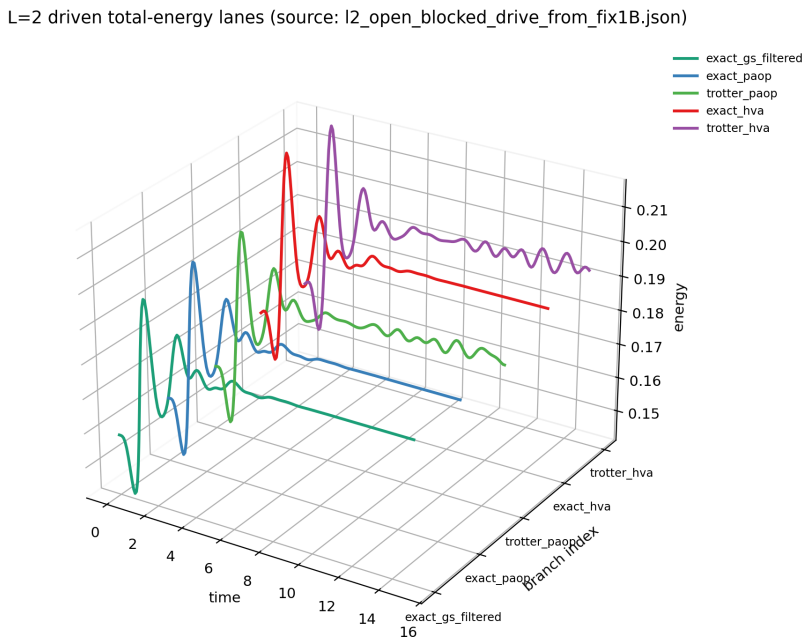

Figure 16: L2 3D driven total-energy lanes

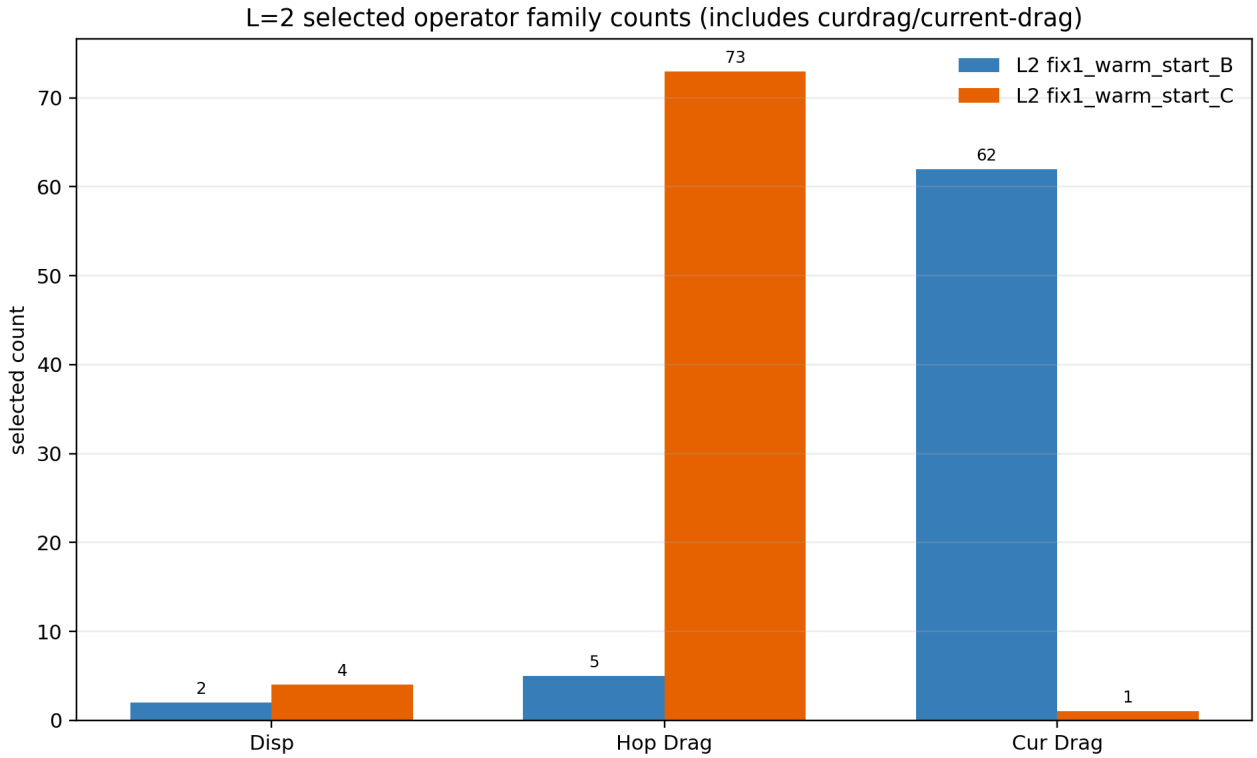

Figure 17: L2 selected operator family counts

11.6 Scale Note (Why driven can look "closer" visually)

L2 quantitative scale check: - No-drive static axis excursion (`E_gs`): ~`2.50e-16` (near-flat line). - Driven static-observable excursion (`E_gs`): ~`8.673e-02`. - Mean gap `|E_gs-E_hva|` no-drive: `8.056e-03`. - Mean gap `|E_gs-E_hva|` driven: `8.942e-03` (slightly larger, not smaller).

Interpretation: visual overlap depends strongly on y-axis excursion; absolute error tables are the oracle for closeness.

Analogous L3 reminder: - L3 static-observable `E_gs` excursion in drive artifact: ~`1.006e-01`. - L3 driven total-energy `E_gs` excursion: ~`5.560e-02`. - Mean `|E_gs-E_hva|` is still large in both views (`~1.856e-01` static-observable, `~1.859e-01` driven-total).

12. Interpretation Notes

1. Large `nfev` does not guarantee lower `DeltaE`; pool geometry and stopping criteria dominate late-stage behavior.
2. The `42 vs 43` depth discrepancy for B is a provenance difference across two executions (export artifact vs accessibility rerun), not a contradiction.
3. In driven outputs, internal HVA branch quality can differ substantially from imported PAOP branch quality; always interpret by branch source first.
4. This report intentionally removes unreadable raw dumps; all raw values remain in JSON artifacts listed in Section 1.